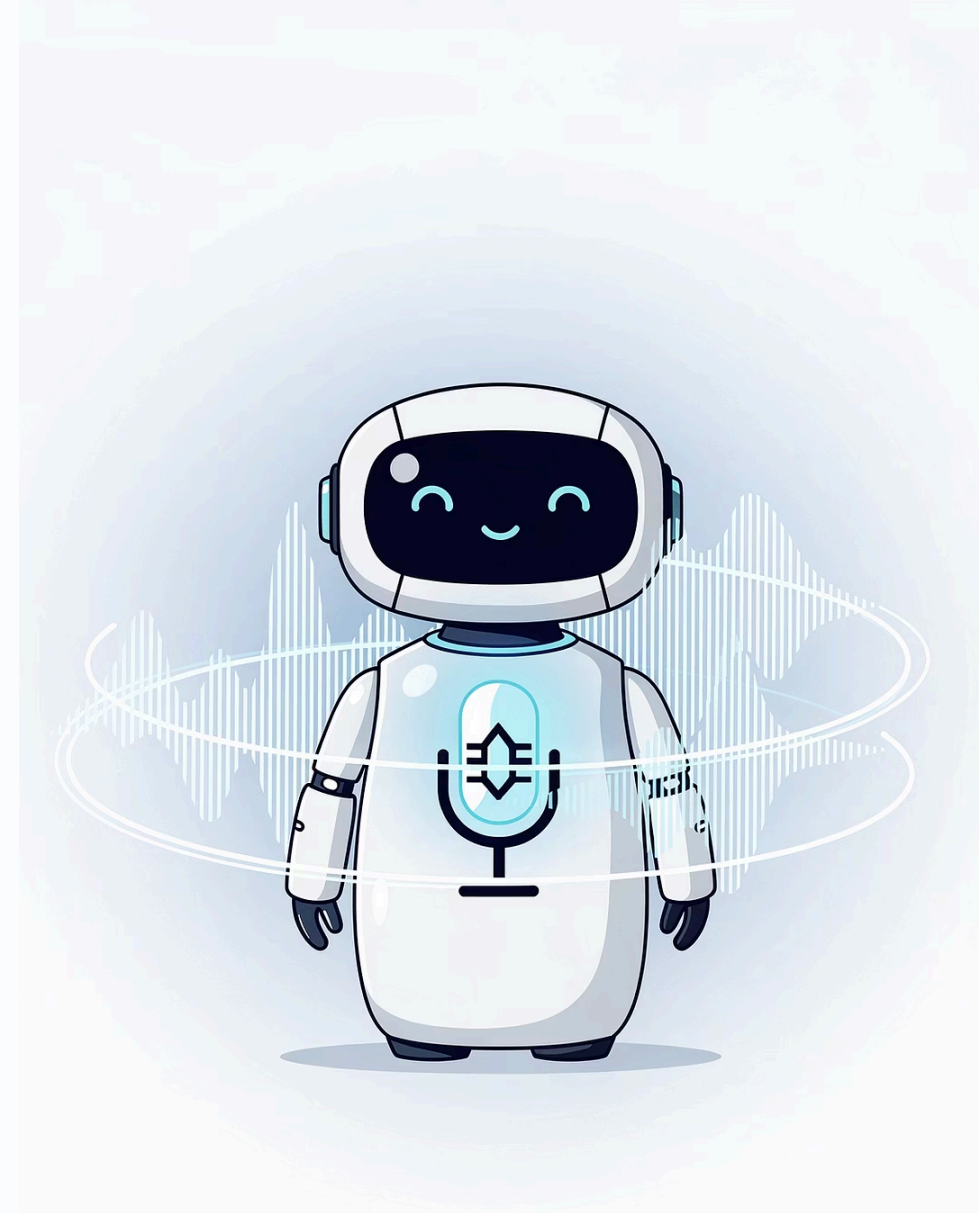


Session 16: AI That Hears & Speaks - Voice AI

You've learned how AI reads. You've learned how AI sees. Today, we discover how AI **HEARS your voice** - and how it can **TALK BACK**.

Plus, you'll get to create **your own AI voice** using a real tool. 🎤



Objectives

By the end of this Session, you'll be able to:

01

Explain Voice AI

Understand what Voice AI is and why it matters

02

Know the 3-Step Recipe

Hear → Process → Speak

03

Sounds into Words

Understand how AI turns sounds into words - and back!

04

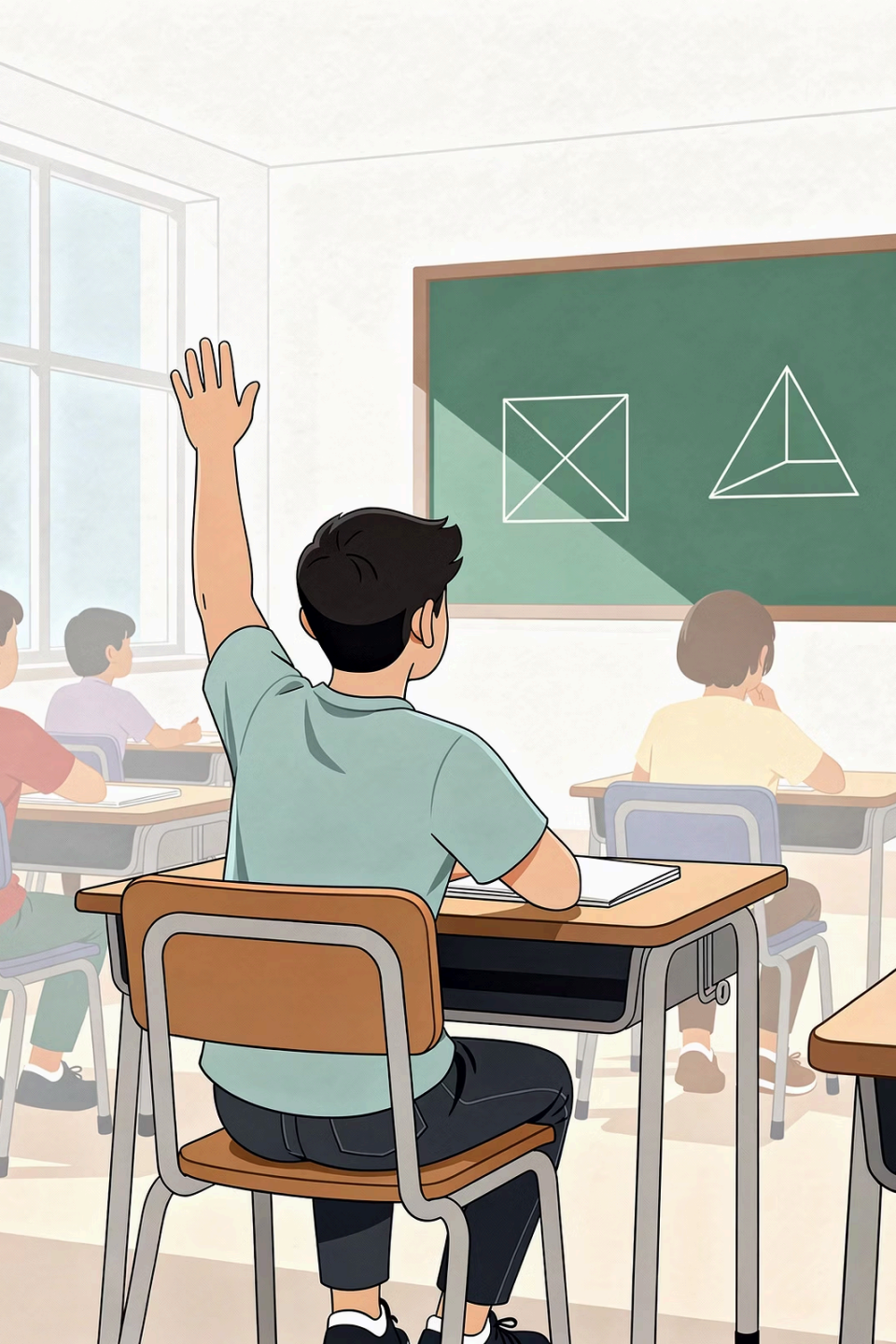
Spot Voice AI Jobs

Identify common Voice AI uses in your everyday life

05

Create Your Own AI Voice

Build your own AI voice using ElevenLabs 



Recap & Homework

Share Your Findings

- Where did you spot Computer Vision this week?
- Did your face unlock challenge reveal anything? *(If you did the bonus)*

Quick Check from Last Week

- What is Computer Vision?
- What's the 3-step recipe? *(See → Process → Act!)*
- What is a pixel?

Warm-Up

Have You Done These?

Said "Hey Siri" or "OK Google"

Talked into your phone to send a text

Asked Alexa to play a song

Used voice chat in a video game

Listened to an audiobook that sounded like a real person

❓ How does a computer turn your voice into words? And how does it talk back?

What is Voice AI?

Voice AI = AI that works with SOUND - especially your voice.

Hearing Words When you talk to it	Understanding What you mean	Speaking Back When it answers
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Voice AI is everywhere:



Voice Assistants

Siri, Alexa, Google Assistant



Phone Bots

Customer service phone bots



Audiobook Narrators

AI-generated voices reading books



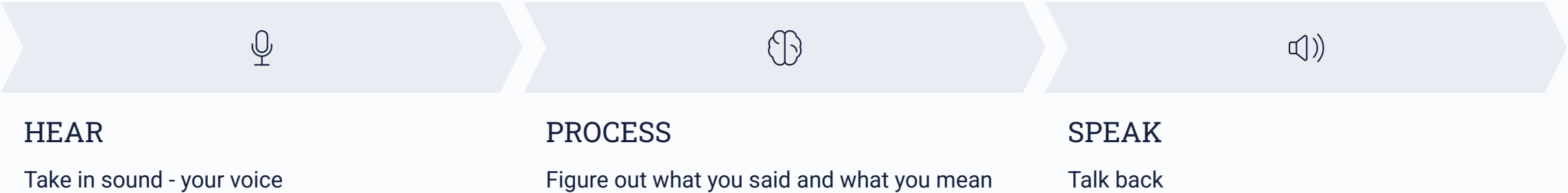
Accessibility Tools

Helping people who can't see or speak



The 3-Step Voice AI Recipe

Most Voice AI follows 3 simple steps:



Notice the pattern? Module 5 Recipes all follow the same structure:

Recipe	Steps
📖 NLP	Break → Understand → Generate
👁 Vision	See → Process → Act
🎤 Voice AI	Hear → Process → Speak

Different senses → same kind of recipe.

Step 1 - HEAR

To AI, your voice isn't really "words." It's a wave of sound.

How It Works

When you talk, you make **vibrations** in the air. A microphone catches those vibrations and turns them into a **squiggly line** called a **waveform**.

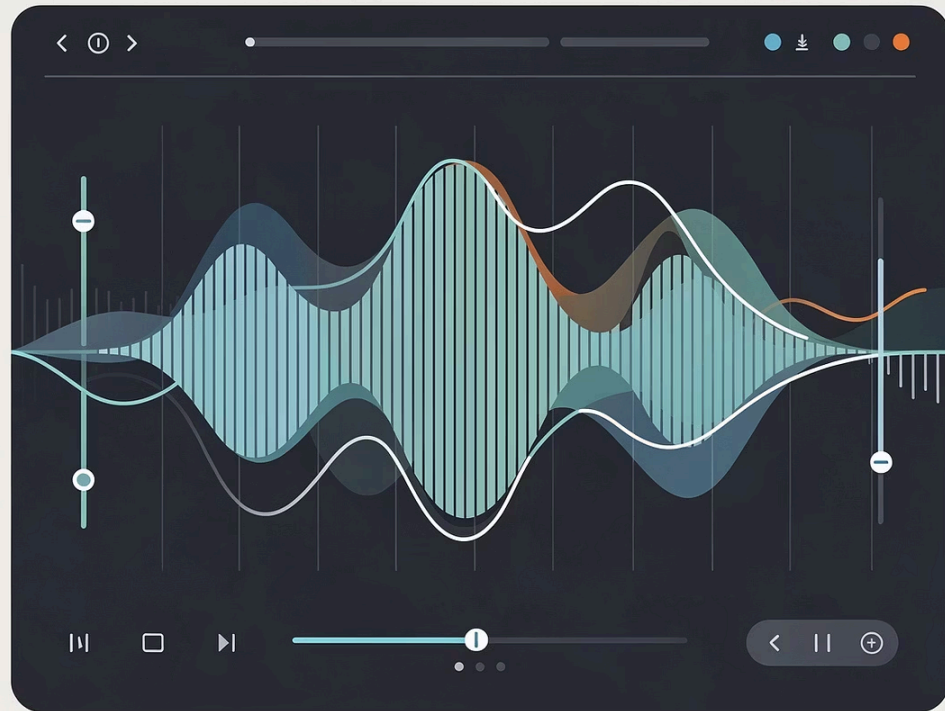
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That squiggly wave = what you said, but in computer language.

Fun Fact 🤔

Different voices make different shaped waves

- High-pitched voices = tighter, faster waves
- Deep voices = wider, slower waves
- Every person's voice is unique in its waveform



From Sound Wave to Words

The waveform is just sound. AI needs to turn it into WORDS.

This step is called **Speech-to-Text (STT)** or **Automatic Speech Recognition (ASR)** - just a fancy name for: **"Turn sound into text."**



You Speak

 *"Hey Siri, set a timer for 5 minutes."*



AI Processes

The sound waveform is analyzed and decoded by the STT engine.



Words Appear

 *"Hey Siri, set a timer for 5 minutes."*

That's STT! AI turned your voice → into words.

Step 2 - PROCESS

Now AI has the words. Time to figure out what you **MEAN**.

The brain part of Voice AI is **NLP**

 What do you want?

Set a timer

 What are the details?

5 minutes

 What action to take?

Start a timer for 5 minutes

Once AI knows what you want, it can take action OR speak back.

Mini Activity 1 - Be the Voice AI! 🎮

Pretend YOU are the Voice AI. For each request, figure out: 🎯 What does the person WANT? ⌚ What are the DETAILS? 🎬 What ACTION should you take?

Person says...	What they want	Details	Action
"Set an alarm for 7am tomorrow."	Set alarm	7am, tomorrow	Open clock app, add alarm
"Play 'New Rules' by Dua Lipa."	?	?	?
"How's the weather today?"	?	?	?
"Text Mom I'll be late."	?	?	?
"Turn off the lights."	?	?	?

Mini Activity 1 - Answers

Person says...	What they want	Details	Action
"Set Alarm for 7am tomorrow."	Set Alarm	7am, tomorrow	Open clock app, add alarm
"Play 'New Rules' by Dua Lipa."	Play music	Song = New Rules, Artist = Dua Lipa	Open music app, play song
"How's the weather today?"	Get weather	Today, your location	Look up weather, speak it back
"Text Mom I'll be late."	Send a text	To = Mom, Message = "I'll be late"	Open messages, send text
"Turn off the lights."	Control a device	Device = lights, Action = off	Send command to smart lights

Step 3 - SPEAK

Finally - AI talks back.

It takes its answer and turns text → into speech. This step is called **Text-to-Speech (TTS)** - just a fancy name for: **"Turn text into spoken voice."**



 **Text Answer**

"Your timer is set for 5 minutes."



 **TTS Engine**

AI converts written words into spoken audio



 **Spoken Voice**

"Your timer is set for 5 minutes." - in Siri's voice

Hear → Process → Speak. The full Voice AI recipe! 



The Two Sides of Voice AI

Voice AI is REALLY two skills working together:



Voice → Text

Speech-to-Text (STT) or ASR (Automatic Speech Recognition)

Converts spoken words into written text so the AI can understand what you said.



Text → Voice

Text-to-Speech (TTS) or Speech Synthesis

Converts written text back into spoken audio so the AI can talk back to you.

You Talk

Your voice is captured by the microphone

It Listens (STT)

Speech-to-Text converts your words into text

It Thinks

The AI processes and generates a response

It Talks Back (TTS)

Text-to-Speech delivers the answer as voice

📌 **Most Voice AI uses BOTH.** You talk → it listens (STT). It thinks → it talks back (TTS).

Common Voice AI Jobs

Voice AI does TONS of jobs. The big ones - all built on Hear → Process → Speak:



Voice Assistant

Listens & responds - Siri, Alexa, Google Assistant



Voice Typing

Speak → text appears. Dictation on your phone.



Voice Narration

Reads text out loud - audiobooks, screen readers



Phone Bots

Customer service AI - "Hi, what can I help you with today?"



Voice Cloning

Makes a voice sound like someone - YouTube videos, AI dubbing



Accessibility

Helps people with disabilities - talk-to-text for someone who can't type

Mini Activity 2 - Match the Voice AI Job!

Look at each example. What Voice AI job is happening?

What's happening	Voice AI Job?
You say "Hey Siri, what time is it?" and Siri tells you	?
You use voice typing to text your friend	?
An audiobook is read by an AI voice that sounds human	?
A YouTuber uses AI to make their video in Spanish - in their own voice	?
A blind person uses AI to read web pages aloud	?
You call the bank and a robot voice helps you	?

 **Options:** Voice Assistant / Voice Typing / Voice Narration / Voice Cloning / Phone Bot / Accessibility

Mini Activity 2 - Answers

What's happening	Voice AI Job
Hey Siri, what time is it?	 Voice Assistant
Voice typing a text	 Voice Typing (uses STT)
AI audiobook narrator	 Voice Narration (uses TTS)
YouTuber in Spanish with their own voice	 Voice Cloning
Reading web pages for blind person	 Accessibility (uses TTS)
Bank robot voice	 Phone Bot

✔ See how Voice AI is everywhere - and helping a lot of different people?

Why Voice AI Is HARD

Even with the recipe, AI struggles with:

Different Accents

AI works better for accents it heard a LOT during training

Bad Audio Quality

Background noise, bad mic - all make it harder

Different Voices

Kids' voices vs adults, men vs women, different pitches

Other Languages

Works better for English than for less-common languages

Mumbling or Speaking Too Fast

Even humans struggle here!

Sarcasm

Same problem as in NLP (Session 14)!

 **Notice something?** AI works WORSE for some voices than others. That's not random - it's because of WHAT data the AI learned from. We'll dig into why this matters in Session 20.



A Big Warning

Voice Cloning Can Be Risky

Voice cloning means AI can copy a real person's voice - using just a few seconds of audio.

The Cool Side

- Audiobooks in any voice
- Helping people who lost their voice
- Translating videos so YouTubers can reach more people

The Scary Side

- Scammers fake a parent's voice to trick kids
- Fake news clips that sound real
- Pretending to be someone famous

⊗  **Real-world rule:** If you get a phone call from someone who sounds like a parent asking for money - **call them back at a number you trust** to make sure it's really them.

Hands-On Activity - Create YOUR Own AI Voice! 🎤

Time for the fun part. Now we'll use a real Voice AI tool called ElevenLabs.

We're going to **design our own AI voice** - just by describing it. Then we'll have it **say something out loud!**

01

Go to ElevenLabs

Visit **elevenlabs.io** in your browser

02

Use Voice Design

Describe your voice in words using the Voice Design tool

03

Listen to What AI Creates

Click Generate and hear your AI voice come to life

04

Make It Say Something Fun

Type a sentence and let your AI voice speak it!



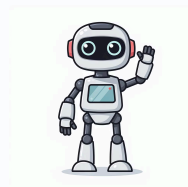
Describe Your Voice 🎨

You'll type a description, and AI creates a voice from it. Try one of these - or make your own!



🧙 The Wise Wizard

"A wise, ancient wizard with a deep, calm voice."



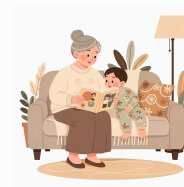
🤖 The Future Robot

"A friendly robot from the future, fast and excited."



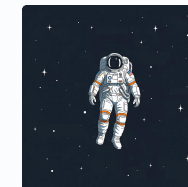
🐉 The Grumpy Dragon

"A grumpy dragon with a low growly voice."



👵 The Kind Grandmother

"A kind grandmother telling a bedtime story, soft and warm."



🚀 The Space Astronaut

"An astronaut narrating their adventure in space."

👉 Click Generate. Then LISTEN. The AI made a voice from your description!



Make Your Voice Speak!

Now make it say something. Go to [Text-to-Speech in ElevenLabs](#). Pick your AI voice. Type a sentence.

Click Generate. Try these:

"Hello! This is my AI voice. Pretty cool, right?"

"Welcome to my podcast about dinosaurs and tacos."

"Once upon a time, in a galaxy far far away..."

"This is the news report at six o'clock."

☒ Listen to your AI voice talk!

Reflection - What Did You Notice?

Discussion:

- Could you tell it was AI? Or did it sound human?
- How close was the voice to what you imagined?
- What did you say with it?
- Where would you USE an AI voice?
- Where could AI voice be MISUSED?
-

 **Voice AI is powerful. With great power comes great responsibility.**

What We Learned in this Session

Voice AI = AI that hears voices and speaks back

The 3-step recipe: Hear → Process → Speak

STT (Speech-to-Text) = voice → words

TTS (Text-to-Speech) = words → voice

Voice cloning is powerful but risky - be careful!

Made your OWN AI voice with ElevenLabs! 

Your Homework - Voice AI Detective

Spot Voice AI in your daily life this week. Find 3 examples of Voice AI being used somewhere around you.

Examples to look for:

Siri / Alexa / Google	Voice Typing	Automated Phone System
AI Voice in YouTube	Read-Aloud Feature	Voice in Video Games
What	Voice AI Job	STT or TTS or Both?

❓ Did any Voice AI mess up? Mishear you? What happened?

Next Time: Session 17

AI That Combines Everything - Multimodal AI

You've seen AI READ. You've seen AI SEE. You've seen AI HEAR. But what if AI could do **ALL of them at once?**

Video + Words

AI that watches a video AND describes it in words

Words → Images → Videos

AI that turns words into images into videos

All Senses Together

AI that hears AND sees AND speaks - all at once

Real Tools

Veo 3 (text → video), **Gamma** (slides from a prompt)

